



Improving the detection of evoked responses to periodic stimulation by using bivariate local spectral F -test – Application to EEG during photic stimulation



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ABSTRACT

The spectral local F -test has been applied for detecting evoked responses to rhythmic stimulation that are embedded in the ongoing electroencephalogram (EEG). Based on the sampling distribution of a flat spectrum at the neighbourhood of the stimulation frequency, spectral peaks in an EEG signal that are due to the stimulation may be readily assessed. Nevertheless, the performance of the technique is strongly affected by both the signal-to-noise ratio (SNR) of the responses and the number of data segments used in the estimation. The present work aims at both deriving and evaluating a multivariate extension of local F -test by including the EEG collected at a second distinct derivation. The detection rate with this multivariate detector was found to be greater than that using a single channel in case of equal SNR in both signals. Monte Carlo simulation results showed that the probability of detection with this new detector saturates for signal-to-noise ratios above 12 dB and indicated a greater detection rate in practical situations, even when smaller SNR-values are found in the added signal (e.g. 5 dB for 16 neighbouring frequencies used in the estimation). The technique was next applied to the EEG from 12 subjects during intermittent, photic stimulation leading to superior performance in comparison with the univariate local F -test. Since a higher detection rate with the proposed technique is achieved without the need of increasing the number of data segments, it allows evoked responses to be detected faster, once the same detection rate may be accomplished with less segments. This might be useful in clinical practice.

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1. Introduction

The spectral F -test has been widely used for automatic detection of periodic signals in noisy environments [1,2], and its definition dates back to seminal works such as those by Rayleigh [3], Schuster [4], Fisher, [5], and Siegel, [6]. Basically, the spectral F -test compares the energy at a given frequency (called signal frequency) with that of the neighbouring frequencies (called noise frequencies). This kind of comparison is often used in clinical practice for detecting, for example, auditory responses in the brain [1,2]. Due to its construction, this test is also called Local Spectral F -test [7,8], which differs from another spectral test that compares the energy of two distinct signal segments [8]. One of the main advantages of

this method is its robustness, since the detection threshold is independent of the signal-to-noise ratio (SNR), which facilitates the development of automatic detection methods.

The local spectral F -test is an objective response detection (ORD) tool. It is based on the comparison of the detector estimate with a statistical threshold that is found based on the detector sampling distribution under the null hypothesis of absence of periodic response. Thus, the detection is achieved when the value of the detector exceeds the threshold value, with constant false positive probability that is equal to the significance level of the test [9]. According to Miranda de Sá and Felix [10], the probability of correctly detecting a response is directly proportional to the SNR and to the total data length used in the estimation of the ORD technique.

In the field of evoked responses detection in the electroencephalogram (EEG), a long stretch of signal may violate the stationarity conditions required for signal processing techniques. Thus, it

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is necessary to develop alternative ways of improving the performance of ORD detectors without increasing the duration of the exam.

An approach that has achieved remarkable results is the Multivariate Objective Response Detection (MORD) [11]. The techniques of this class use more than one signal for detection, starting from the premise that the response to be detected spreads to several different channels and that the background noises of each channel are not correlated [12]. Miranda de Sá and Felix [10] extended the concepts of the univariate detector magnitude-squared coherence [13] to the bi-variate case and showed that the use of two channels improves the detection ratio of evoked responses.

In this work, the local spectral F -test is extended to the multivariate case using the sum of two individual F -tests. The distribution under the null hypothesis of flat spectrum at the neighbourhood of the signal frequency was obtained theoretically for this bivariate detector, as well as the performance of the test was assessed by means of Monte Carlo simulation. The new detector was then applied to the detection of evoked responses in the electroencephalogram (EEG) during intermittent photic stimulation.

2. The spectral local F -test – $F_y(f_0)$

Considering a linear model with an additive noise parcel in the output, the spectral local F -test applied to a discrete-time signal $y[k]$ is defined as the ratio between the power spectrum estimate at frequency f_0 and the average power spectrum estimate in the L neighbouring frequencies to f_0 [1], i.e.

$$F_y(f_0) = \frac{|Y(f_0)|^2}{\frac{1}{L} \sum_{i=0-L/2}^{0+L/2} |Y(f_i)|^2}, \quad (1)$$

where $Y(f_0)$ is the discrete Fourier Transform of $y[k]$ at frequency f_0 and $Y(f_i)$ are those values at the even number of closest neighbouring frequencies (L) to f_0 . Assuming a flat (white) spectrum around f_0 , this statistic may be used for assessing if the power spectrum at f_0 is statistically different from its neighbours.

For the null-hypothesis (H_0) of no evoked response and considering $y[k]$ as Gaussian distributed, the sampling distribution of $F_y(f_0)$ is given as (after [1]):

$$F_y(f_0)|_{H_0} \sim F_{2,2L}, \quad (2)$$

Where the symbol “ $\sim$ ” means “is distributed as” and $F_{2,2L}$ is the Fisher–Snedecor distribution with 2 and $2L$ degrees of freedom. The definite integral of the probability density function in (2) allows obtaining the critical value according to:

$$\int_0^{F_y(f_0)_{crit}} p_{F_y(f_0)}(z) dz = \int_0^{F_y(f_0)_{crit}} \left(1 + \frac{1}{L}z\right)^{-(L+1)} dz = 1 - \alpha, \quad (3)$$

where $F_y(f_0)_{crit}$ is the critical value for a given significance level (α). After considerable algebraic simplification, the critical values for the F -test are given as

$$F_y(f_0)_{crit} = L \left(\alpha^{-\frac{1}{L+1}} - 1 \right). \quad (4)$$

For the alternative hypothesis (H_1) of presence of evoked responses, the local F -test distribution becomes the non-central Fisher–Snedecor distribution with 2 and $2L$ degrees of freedom and non-centrality parameter equal to $2 \cdot \text{SNR}$ [12]

$$F_y(f_0)|_{H_1} \sim F'_{2,2L}(2 \cdot \text{SNR}) \quad (5)$$

3. Methods

3.1. New method: extension to the multivariate case – $F_{y2y1}(f)$

The $F_y(f_0)$ belongs to the class of constant-false alarm ratio (CFAR) detectors, whose main characteristic lies in the fact that

their false positive ratio is constant and equal to the significance level of the test [9]. The power of the test is proportional to the number of degrees of freedom of the detector, which in this case is related to L . However, Picton et al. [1] stated that $L > 20$ does not lead to significant improvement in the performance of $F_y(f_0)$. Alternatively, the probability of detecting a response increases if the SNR is higher. This can be achieved by coherently averaging many data sweeps prior to application of $F_y(f_0)$. However, collecting more EEG data may not be appropriated in some situations such as intra-surgical monitoring and paediatric audiometric tests and therefore improvements in the detection power must be achieved for a fixed SNR.

It can be stated that the cerebral response evoked by a stimulus spreads out to more than one EEG derivation at the same time [14]. Thus, we propose a new detector, $F_{y2y1}(f_0)$, which is the sum of two independent $F_y(f_0)$ taken from two different signals. It is assumed that each signal has a portion of the response and that the background noises of the signals are uncorrelated. The bivariate local F -test can then be written in an equation form as follows:

$$F_{y2y1}(f_0) = F_{y1}(f_0) + F_{y2}(f_0) \quad (6)$$

where $F_{y1}(f_0)$ and $F_{y2}(f_0)$ are similar local F -tests applied to signals $y_1[k]$ and $y_2[k]$, respectively.

Since $F_{y2y1}(f_0)$ is the sum of two Fisher–Snedecor-distributed variables, the distribution under H_0 is the convolution of those two F distributions. Although such a convolution is not a straightforward task and is still an open problem, Dyer [15] obtained the cumulative density function (CDF) of the sum of two generalized F -distributions and this result can be used to obtain the critical values for $F_{y2y1}(f_0)$.

By adjusting the expression derived by Dyer [15] (Eq. (2.11), specifically), the theoretical expression of the CDF of $F_{y2y1}(f_0)$ under H_0 is:

$$\begin{aligned} \text{CDF}(F_{y2y1}(f_0)) &= \sum_{j=0}^{L-1} \sum_{k=0}^{L-1} (-1)^{j+k} C_{j,k} \\ &\quad \times {}_2F_1[j+1, k+1; j+k+3; 1-2(1-\theta)] \end{aligned} \quad (7)$$

where $\theta = \frac{1}{L} F_{y2y1}(f_0) / (1 + \frac{1}{L} F_{y2y1}(f_0))$, ${}_2F_1[\bullet]$ is the Gauss's hypergeometric function of one variable and

$$C_{j,k} = e^{2 \ln \Gamma(L+1) + 2 \ln \Gamma(k+1) + \ln \Gamma(j+k+3) - 2 \ln \Gamma(L-j) + \ln \Gamma(k+1) + (j+1) \ln \phi + (k+1) \ln \phi}, \quad (8)$$

where $\Gamma(\bullet)$ is the gamma function.

3.2. Simulation studies

The probability of detecting a response as a function of the SNR is related to the distribution of the detector under hypothesis H_1 , which would require the convolution of two non-central F distributions, leading hence to a more complex expression. Alternatively, this probability of detection (PD) may be obtained by using a Monte Carlo simulation. Based on the latter approach, 10,000 pairs of simulated signals for each SNR-value were generated to estimate the probability of detection for $F_{y2y1}(f_0)$ as a function of the SNR and for increasing L -values (the reader is referred to Ref. [10] for more details on this Monte Carlo simulation).

Another Monte Carlo simulation was performed in order to assess the probability of mistakenly detecting a response with $F_{y2y1}(f_0)$. In this case, 10,000 pairs of white Gaussian noise were generated and $F_{y2y1}(f_0)$ was estimated for each pair. Next, the false positive ratio was obtained by counting the cases in which $F_{y2y1}(f_0)$ was greater than the critical value and dividing by the total number of tests. The significance level of test was 5%.

The PD is obtained under the assumption that the two neighbouring signals are mutually independent. However, uncorrelated

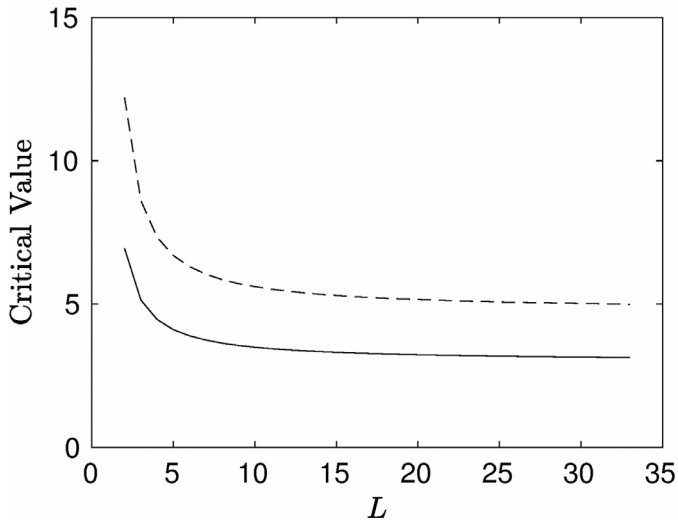


Fig. 1. Theoretical critical values for $F_{y_{2y1}}(f_o)$ (dashed line) and $F_y(f_o)$ (continuous line) as a function of L .

background activities are not likely to occur in practical cases. Thus, an investigation of such effect was conducted by including different correlation levels in addition to the PD simulation. The SNR was kept fixed. This simulation is more detailed in Felix et al. [11].

3.3. EEG data acquisition

In order to evaluate the performance of $F_{y_{2y1}}(f_o)$ and compare it with that of $F_y(f_o)$, the EEG signal of 12 normal young subjects (age range: 9–17 years) was recorded over a period of 16 s during intermittent photic (flash) stimulation (IPS) at 6 Hz. The signals at the derivations O1 and O2 (reference: ipsilateral earlobe) were recorded and digitized at 256 Hz. The choice of such derivations was based on the fact that they lie above the visual cortex, and hence the evoked responses are expected to be stronger at these locations.

4. Results and discussion

4.1. Critical values

Theoretical critical values for $F_{y_{2y1}}(f_o)$ may be obtained from Eq. (7) by taking the abscissa values where the ordinate reaches $(1 - \alpha)$. These critical values are shown in Fig. 1 for increasing values of L and for $\alpha = 0.05$. In order to allow a comparison, the critical values for $F_y(f_o)$, obtained using Eq. (4), are also shown.

Corresponding, the critical values for the multivariate technique are always higher than the according univariate ones and this can be explained by the increase in the degrees of freedom in the estimation of $F_{y_{2y1}}(f_o)$.

4.2. Probability of detection

The probability of detection with $F_{y_{2y1}}(f_o)$ is shown in Fig. 2. In order to allow comparison, the same simulation was performed for $F_y(f_o)$ and both results are shown in Fig. 2.

As expected, the probability of detecting a response is a monotonically increasing function of the SNR [9,10]. Moreover, PD with both detectors can also be augmented by increasing L up to 20 as suggested in Picton et al. [1]. However, an improvement in the detection is expected for a fixed SNR by using $F_{y_{2y1}}(f_o)$ instead of $F_y(f_o)$, since for SNR-values greater than -4 dB PD-values

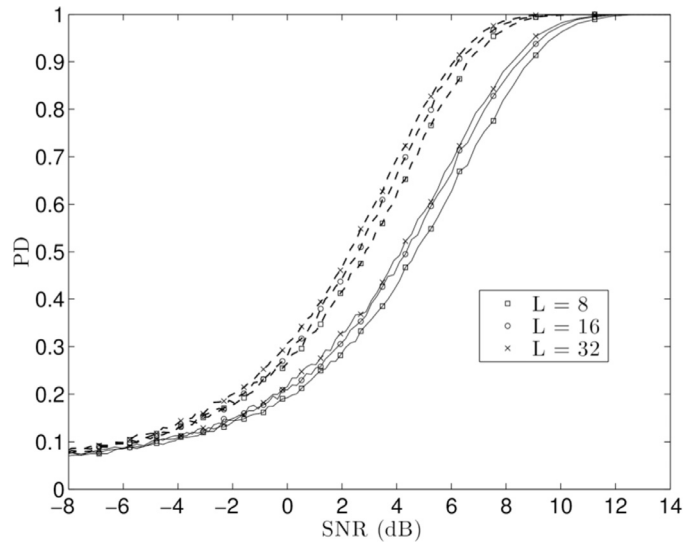


Fig. 2. Curves for PD of responses using $F_{y_{2y1}}(f_o)$ (dashed lines) as a function of SNR and for different L -values. PD using $F_y(f_o)$ is also shown as continuous lines.

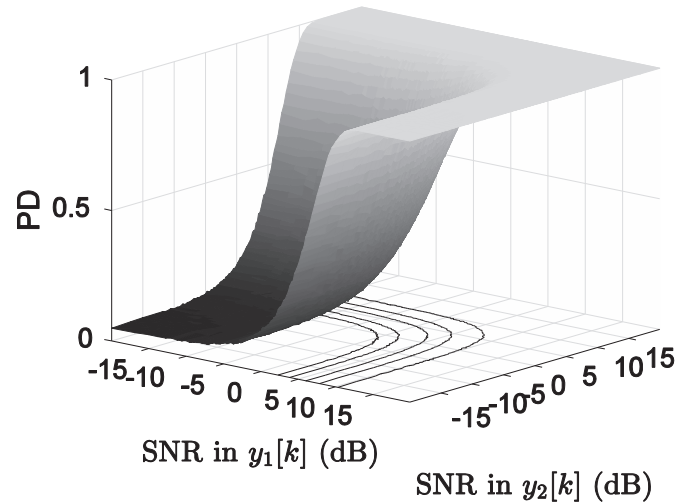


Fig. 3. Probability of detection with $F_{y_{2y1}}(f_o)$ for different SNRs in $y_1[k]$ and $y_2[k]$ for $L = 16$. The curves in the xy -plane are contour curves for $PD = 0.25, 0.35, 0.55, 0.75$ and 0.95 .

with $F_{y_{2y1}}(f_o)$ are consistently greater than the according ones with $F_y(f_o)$. Thus, it is always better using $F_{y_{2y1}}(f_o)$ if two EEG signals with the same SNR are available.

Note that the above results apply only for the special case when the SNR is the same for both signals. The general case of different values of SNR for $y_1[k]$ and $y_2[k]$ is now investigated for $L = 16$ and 10,000 iterations. The result is shown in the surface plot of Fig. 3.

The result in Fig. 2 corresponds to the intersection between the bisector plane and the 3D curves such as that of Fig. 3 for $L = 16$ and equal SNR in both signals.

As it can be noted, for a fixed SNR value in $y_1[k]$, an infinite number of PD values may be found depending on the SNR in $y_2[k]$. On the other hand, if the probability of detection is fixed instead, the infinite pairs of SNR in $y_1[k]$ and in $y_2[k]$ leading to the same PD may be found as the contour maps in the xy -plane of Fig. 3. Such information is important in order to obtain the minimum SNR in $y_2[k]$ that improves the detection rate when using both $y_1[k]$ and in $y_2[k]$ [detection with $F_{y_{2y1}}(f_o)$] in comparison with that using only one signal [detection with $F_y(f_o)$].

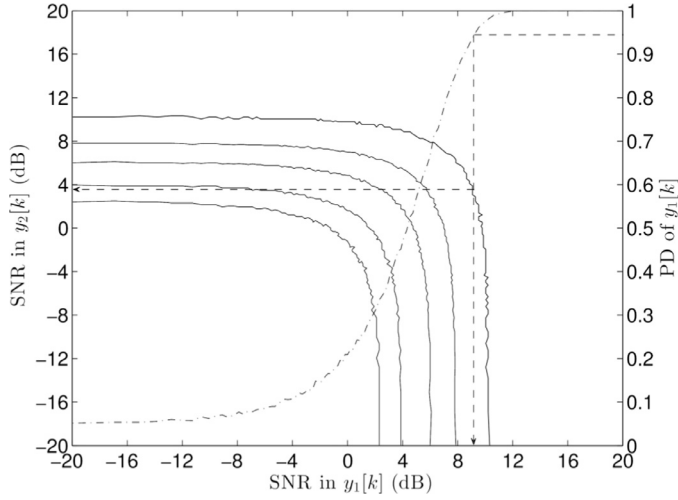


Fig. 4. Contour curves of the 3D-plot shown in Fig. 3 for PD=0.25 to 0.95 in steps of 0.20. PD with a single channel ($y_1[k]$) is also plotted in dashed line with y-axis at right side to show the corresponding SNR in such signal (9 dB) that leads to 95%-detection rate. Based on this value, the threshold of ~4 dB is found for the SNR in $y_2[k]$ that results in detection improvement if both signals are employed.

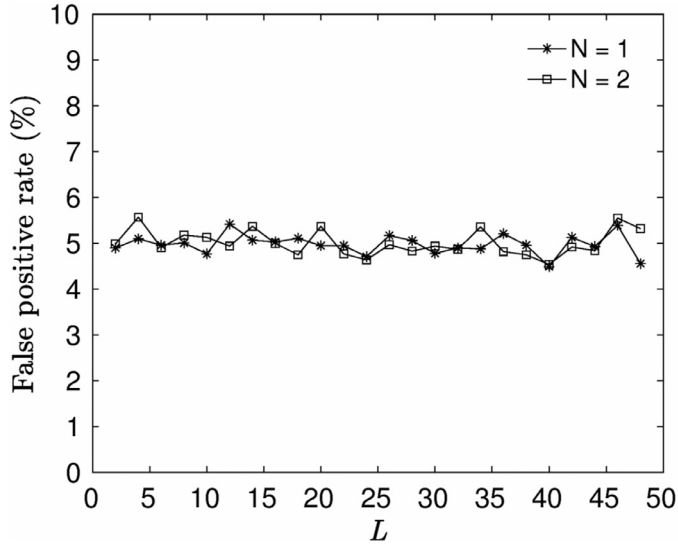


Fig. 5. False alarm ratio for $F_{y2y1}(f_0)$ (with squares) and for $F_y(f_0)$ (with asterisks).

In Fig. 4, the case of PD=0.95 is taken as an example. If only one signal is used in the detection with $F_y(f_0)$, a PD of 0.95 is expected for an SNR of 9 dB with $L=16$ (see Fig. 2). Setting this value of SNR for $y_1[k]$ on the contour curve of Fig. 4, a corresponding SNR of about 4 dB is found for $y_2[k]$. Thus, for SNR values greater than this one, an improvement in the detection is expected, since the intersection between the horizontal and vertical dotted lines would lie on a contour curve above the one for PD=0.95. On the other hand, for an SNR in $y_2[k]$ lower than this threshold, a smaller detection rate is expected when using both signals.

4.3. False alarm ratio

Fig. 5 indicates that the false alarm rates exhibit very small random fluctuations around 5% without a monotonic trend in spite of either including the second signal or increasing the number of neighbouring frequencies. This suggests that the expected false alarm ratio tends to remain equal to the significance level.

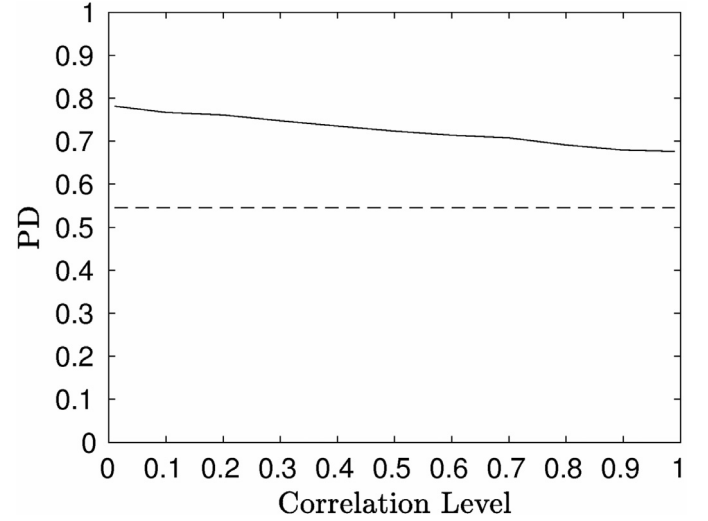


Fig. 6. Probability of detection (PD) with $F_{y2y1}(f_0)$ as a function of the correlation level between the background noises in 5 dB SNR signals (continuous line). PD-value with $F_y(f_0)$ in a single channel with same SNR is shown in dashed line to allow comparison.

Table 1

Detection ratio of all subjects for both detectors.

Subject	$F_y(f_0)$	$F_{y2y1}(f_0)$
#1	84.21	94.74
#2	36.84	68.42
#3	63.16	84.21
#4	26.32	73.68
#5	42.11	68.42
#6	68.42	84.21
#7	42.11	63.16
#8	26.32	84.21
#9	52.63	73.68
#10	36.84	84.21
#11	42.11	73.68
#12	21.05	63.16
Mean	45.18	76.32

4.4. Effect of correlation of the background noise

The PD estimate with $F_{y2y1}(f_0)$ as a function of the correlation level for fixed SNR-value (=5 dB) is shown in Fig. 6. As it can be observed, PD decreases as the correlation level of the background noise increases. However, it always remains greater than the PD with $F_y(f_0)$ in a single channel with the same SNR.

4.5. Application to EEG signals: detection ratio

The 16-s signals were divided into epochs of 2 s each and then averaged. Next, the $F_y(f_0)$ was then calculated following Eq. (1) for O1 and $F_{y2y1}(f_0)$ was calculated using Eq. (6) for both derivations, with $L=16$. The detection ratio was evaluated on the response in all harmonics of the stimulation frequency, with significance level of 5%. The results are shown in Table 1.

The performance of $F_{y2y1}(f_0)$ is always greater than the univariate version and this difference is statistically significant according to the paired t -test ($p < 0.01$).

5. Conclusion

In this work, an objective response detection technique was developed based on the extension of the spectral local F -test to the multivariate case, specifically to the sum of two independent

F-tests. This approach takes into consideration that the evoked response to be detected spreads to more than one EEG derivations.

The critical values for the new detector – $F_{y2y1}(f_o)$ – could be assessed theoretically due to the adaptation of the results found in [15]. The critical values are the statistical thresholds needed to automate the detection protocol of evoked responses, which is important in many areas of biomedical signal processing.

In order to assess the probability of detecting a response, the detector distribution under the alternative hypothesis of response being present must be known. Although the distribution of $F_y(f_o)$ under H_1 is given in Eq. (5), the sum of non-central Fisher–Snedecor distributions is not a straightforward task. To circumvent this problem, Monte Carlo simulations were carried out and it was found that the PD with $F_{y2y1}(f_o)$ is always greater when compared to $F_y(f_o)$ when the SNR is the same in both signals (Fig. 2). Furthermore, it was demonstrated that if the SNR of the signals are different, the SNR of the second signal must be always smaller than the SNR of the first signal, at least to maintain the PD constant (Fig. 4).

Although the probability of detection with $F_{y2y1}(f_o)$ was obtained assuming uncorrelated background activities, the effect of distinct correlation levels was also investigated with simulated data. As expected, the detection rate decreases as the correlation between the noise parcels increases. However, even with perfect correlation in such parcels, the probability of detection with the bi-variate detector remains greater than that using a single channel (Fig. 6).

The improvements in terms of theoretical performance obtained by adding together two independent *F*-tests do not affected the constant-false-alarm-ratio properties of the detector (Fig. 5), since this ratio is kept around 5% for different values of *L*. Thus, $F_{y2y1}(f_o)$ is also a CFAR detector [9].

When applied to the EEG during stroboscopic stimulation, the detection ratio was always higher with $F_{y2y1}(f_o)$ (Table 1). In fact, an average improvement of ~30% was observed when O1 and O2 were analysed together using the new multivariate technique.

Finally, it may be concluded that $F_{y2y1}(f_o)$ is a useful technique for detection of evoked responses in the EEG. The methods developed here are mathematically sound, since the performance predicted by Monte Carlo simulations were observed in the application in real data.

Competing interests

No conflict of interest.

Ethical approval

Work approved by the Local Ethics Committee (Parecer 100/2011; Processo 36/2011 – UFRJ).

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